When Keyword Measures of Thematic Exposure Break Down: Can a Disclosure-Based Evidence Ladder Substitute?

An identification-explicit case study in Japanese equities

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Abstract

Measures of a firm's exposure to an investment theme are commonly built by matching keywords against firm text. We audit one such measure across 3,649 Japanese listed firms and find that it *saturates*: it takes 20 distinct values, and 99.0% of firms sit exactly at their own sector's median. 273 firms — a semiconductor photomask-inspection monopolist, a fire-alarm maker, a watchmaker, a railway-signalling firm — carry an identical score. The mechanism is checkable: the keyword match reads firm metadata rather than disclosure text, a sector bonus is hard-coded on top, and the one continuous firm-level input is populated for 2 of 3,649 firms. Arithmetically the measure is a sector label. We propose a disclosure-graded *evidence ladder* as the alternative and define it as a coding protocol, then turn it on the portfolio this paper studies: of 20 holdings, 1 carries artefact-backed thematic evidence. We believed our own selection was evidence-based; it was not, and we could not tell from the inside. Performance claims are partitioned into three layers and calibrated within them: out-of-sample, the mechanical screen shows no evidence of adding value (15 of 15 specification cells underperform TOPIX) under a design too weak to license the stronger negative claim; the full portfolio sits high in a randomisation distribution and equally high in that distribution's beta and drawdown tails; and its factor alpha, while surviving a sector control, is *smaller* than the alpha of the purely mechanical benchmark built from the same gate. The contribution is a measurement one. Every number is generated by scripts in the replication package, and the 85 specifications we traversed are counted and disclosed.

What this paper does not claim. It does not claim that the portfolio studied here beat any benchmark out-of-sample; Table 4 says the opposite for the part we can test. It does not claim the evidence ladder predicts returns — we have not validated it against realised fundamentals, and we say what such a validation would require. Comparisons to a Buffett-style benchmark appear only as the motivating story that produced the data, and are reported inside the layer that owns them.

1 Motivation

This paper began as an undergraduate stock-selection project and became a measurement paper by accident. We were building a portfolio around exposure to the AI investment theme, and we needed to rank Japanese firms by how exposed they were. We had a composite AI-exposure score of the ordinary kind, so we sorted on it.

The sort did not work. One of our candidate holdings, santec, makes tunable lasers and optical test instruments — a plausible beneficiary of optical interconnect demand in data centres. Its score was

0.9723. So was the score of 273 other firms, among them Lasertec, the sole global supplier of photomask inspection systems for EUV lithography, and also a fire-alarm manufacturer, a watch company, and a railway signalling firm. Only 3 firms in the entire market scored higher than any of them. We could not use the measure to choose between a monopolist in advanced semiconductor metrology and a maker of smoke detectors, because the measure did not distinguish them.

Our first reaction was that we had found a quirk in one index. Investigating the construction (Section 2.1) showed something more systematic, and turning our own proposed remedy back on our own portfolio (Section 2.3) showed that we had been relying on the broken measure without knowing it. That sequence — a measure that cannot rank, a mechanism that explains why, and a study that failed its own test — is what the paper reports. The portfolio's performance appears here only as the vehicle that carried us to the question, reported inside the identification layer that owns it.

2 The measurement problem

2.1 Saturation, and the mechanism behind it

We audit the thematic-exposure score that our own selection pipeline inherited — a composite AI-exposure index of exactly the kind an investor would take off the shelf — across all 3,649 firms in the cross-section. Table 1 reports what it resolves.

The score takes 20 distinct values. Its five largest ties account for 90.2% of all firms, and the effective number of levels, computed as the inverse Herfindahl index of the tie-size shares, is 3.91. 273 firms share the single value 0.9723, and only 3 firms in the entire market score above them. Inside that tie sit Lasertec, the sole global supplier of photomask inspection systems for advanced semiconductor lithography, and — indistinguishably — a fire-alarm manufacturer, a watchmaker, and a railway signalling company. This is not noise around a usable signal. In the region of the distribution where an investor needs the measure to discriminate, it does not order firms at all.

The mechanism is worth stating precisely, because it is specific and checkable rather than a general complaint about keyword methods. Three construction choices compound:

1. **The keyword match never reads a disclosure.** It runs over firm metadata — company name, the 33- and 17-sector labels, market segment, and scale category. A firm can reach a theme bucket without ever having mentioned the theme, and a firm that discusses it at length in its filings gains nothing.
2. **A hard-coded sector bonus is added.** Membership in Electric Appliances, Machinery, Chemicals, Nonferrous Metals, Metal Products, or Precision Instruments adds a fixed increment to the AI-infrastructure bucket, with parallel bonuses for the other buckets. Since the sector label is also one of the matched text fields, sector enters twice.
3. **The only continuous firm-level input is missing and silently imputed.** R&D intensity carries a quarter of the composite's weight. It is populated for 2 of 3,649 firms and filled with zero for the remaining 3,647.

Together these leave nothing that varies across firms within a sector. The arithmetic consequence is visible in the data: 99.0% of firms sit exactly at their sector's median score, and 18 of 33 sectors are single-valued — every firm in them receives an identical score. **The measure is a sector label presented as a firm-level exposure estimate.** We emphasise that we found this by auditing a measure we had ourselves relied on, and that the failure is not exotic: each of the three choices above is individually defensible, and a user who never inspects the missingness of one input would not see the result.

We take the general lesson to be about the direction of the field's remedy rather than about this one index. Hoberg and Phillips [2016] construct industry classifications from firm-written product descriptions precisely because assigned sector codes lose firm-specific information; a thematic measure that reduces to a sector code has thrown away the same information while appearing not to. The diagnostic is cheap and, we would argue, should be routine: report the number of distinct values a proposed exposure measure takes, and the share of firms sitting at their sector's median.

Table 1: Measurement audit (no inference layer). *Saturation of a thematic-exposure measure.* This table makes no claim about returns. All 3,649 listed firms in the cross-section, scored by the composite AI-exposure index inherited by our own pipeline.

Resolution	
Firms scored	3,649
Distinct score values	20
Effective number of levels (inverse Herfindahl)	3.91
Five largest ties, share of cross-section	90.2%
Firms exactly at their sector's median score	99.0%
Single-valued sectors	18 of 33
The largest tie at the top: 273 firms at a score of 0.9723, with 3 firms above	
Lasertec Corporation (*semiconductor photomask inspection; sole global supplier*)	—
HOCHIKI CORPORATION (*fire alarms and detection equipment*)	—
Citizen Watch Co.,Ltd. (*wristwatches*)	—
Nippon Signal Company,Limited (*railway signalling*)	—
santec Holdings Corporation (*optical test instruments for fibre networks*)	—
Mechanism	
Keyword match reads disclosure text?	**No**
Fields matched	metadata only[a]
R&D intensity (weight 0.25) populated for	2 of 3,649 firms

[a] Company name, 33-sector label, 17-sector label, market segment, scale category; plus a hard-coded additive bonus for membership in designated sectors. Since the sector label is both a matched field and a bonus trigger, sector enters twice, and no remaining input varies across firms within a sector.

2.2 An evidence ladder, defined so it can be re-coded

If keyword presence cannot order firms, the alternative is to grade what a firm's disclosures actually substantiate. We define three levels, stated as a coding protocol rather than a concept, and report their distribution over the 1,200-firm candidate pool in Table 2.

Level 1 — keyword only. A keyword places the firm in a theme bucket, and no retrievable artefact names a product, a customer, or a committed investment. Assigned mechanically.

Level 2 — named product or customer. A retrievable artefact (a filing passage, product page, or IR document) names a specific product, customer, or committed investment tied to the theme. The coder records the source URL, the source type, and the quoted passage.

Level 3 — disclosed quantity. The artefact carries a number attributable to the theme: revenue, order book, unit or customer count, or capital expenditure, with its as-of date. The number and the quotation are both recorded.

The distribution is the point. Of 1,200 firms, 600 have no thematic link at all and 586 are level 1 — keyword-reachable and nothing more. Only 8 reach level 2 and 6 reach level 3. The ladder's discriminating power is concentrated in the 14 firms where a human opened a document; everything else is the saturated region of Section 2.1 under a different name. Every level-2 and level-3 assignment in this study carries a URL and a quoted snippet, so a referee can re-open the artefact and disagree with the coding. Levels 0 and 1 require no judgement and are reproducible by script.

Table 2: Measurement audit (no inference layer). *The evidence ladder, and what our own portfolio scores on it.* Top-1200 candidate pool from the contest edition's value-quality screen. Levels 0 and 1 are assigned by script; levels 2 and 3 each carry a source URL, source type, and quoted passage.

Level	Requirement	Firms	Share
Panel A: the candidate pool			
0	No thematic keyword link	600	50.0%
1	Keyword hit only; no artefact	586	48.8%
2	Named product, customer, or committed investment	8	0.7%
3	Disclosed quantity attributable to the theme	6	0.5%
Verified by hand (levels 2–3)		14	1.2%
Discarded as keyword-only		577	
Panel B: the same ladder applied to the 20 holdings this paper studies			
0	No thematic keyword link	5	25.0%
1	Keyword hit only	9	45.0%
2	Named product or customer	1	5.0%
3	Disclosed quantity	0	0.0%
—	*Never scored (outside the pool)*	5	25.0%

Notes. The ladder's discriminating power is concentrated in the 14 firms where a human opened a document. The 577 discarded firms' filings were *not* read; they are the residual after hand verification, not a population that was inspected and rejected. The ladder is **not validated** as a predictor — see Section 2.3 for what a validation would require.

What the 577 figure does and does not mean. Our own earlier write-up reported that 577 firms were excluded as "keyword-only". Stated carelessly this sounds like 577 sets of filings read and found wanting. It is the opposite: it is the residual after 14 firms were verified by hand, discarded as a block. **Those 577 firms' filings were not read.** We correct the phrasing here because the looser version credits the study with work it did not do.

2.3 Applying the ladder to our own portfolio

The obvious test of a proposed measure is to turn it on the study that proposes it. We did that late, and it produced the most uncomfortable result in this paper.

Of the 20 holdings in the portfolio whose performance we report, 5 never entered the candidate pool and so were never assigned a thematic level at all. Among the 15 that were, 1 carries a hand-verified level-2 artefact — Lasertec, with a quoted statement about EUV mask-blank inspection and a source URL. 9 sit at level 1: reachable by the metadata keyword screen and nothing more. 5 sit at level 0, with no thematic keyword link at all. And the portfolio shares only 1 name with the selection pipeline's own final-20 list.

The thematic leg of our own portfolio does not clear the ladder we are proposing. It sits almost entirely inside the saturated region of Section 2.1 — the region we argue is uninformative. The working file that records the role rationales states, at the top, that the business descriptions are drafted from general knowledge and require verification against IR and EDINET before submission; for these fifteen holdings that verification was never completed. We had described the selection in an earlier draft as reading disclosures. It was not.

We report this for three reasons. It is true. It changes what Layer 2 can mean: the randomisation result in Section 4.2 compares a discretionary portfolio against random draws, but the discretion was narrative rather than evidentiary, so "selection skill" is not even the right name for what is being tested. And it relocates this paper's motivation honestly. The evidence ladder is not proposed because other people's measures are poor. It is proposed because our own selection process, which we believed was evidence-based, turns out on inspection to have been keyword-and-story based for nineteen of twenty holdings — and we could not tell from the inside. That is the failure mode we think is general, and it is the reason the diagnostic in Section 2.1 should be routine rather than exceptional.

We have not validated the ladder. Nothing above shows the ladder predicts anything. It is a measurement proposal with a coding protocol, not a tested signal, and the honest description of its status is that it survived only the test of being cheaper to disagree with than a keyword count. A validation would need, at minimum: coding the ladder at an as-of date from filings available at that date, on a sample large enough for a cross-sectional test; a pre-specified outcome, for which realised theme-attributable revenue growth is the natural choice and is disclosed by only a minority of firms; inter-coder agreement statistics, since levels 2 and 3 require judgement that 14 hand assignments cannot establish; and a head-to-head comparison against the saturated measure on the same sample and horizon.

The crudest possible version of that test is in Appendix Table 12, and it does not favour us: among the 14 hand-coded firms, those at level 3 grew revenue *more slowly* over the window than those at level 2. We report it because a paper arguing for careful measurement cannot omit the one descriptive check it can compute merely because the check is unflattering — but we would resist any reading of it, in either direction, for the reasons given in the table's notes. Designing the study that could actually order the levels is the question we would most like a reader's view on, and it is the second question in our cover letter.

3 Data and identification

3.1 The point-in-time panel

Our accounting data is a panel of annual securities reports filed with EDINET: 10,712 filings covering 3,579 firms, a median of 3 fiscal years each. Each filing carries its *submission timestamp* alongside the fiscal period it covers, and that single field is what makes Layer 1 possible. It lets us reconstruct, for any date, exactly which firms' financials a screen could have seen — a distinction that a panel indexed only by fiscal period silently erases, because a fiscal year ending in March is not public until it is filed, typically three months later. The funnel in Appendix Table 9 reports how much of the cross-section is visible at each of our as-of dates: at 2024-10-01, 87 firms clear the quality gate out of the 2,224 firms with enough filing history to screen, and most of the attrition is firms with only one fiscal year on file rather than firms failing the gate.

Prices are daily adjusted closes for 3,650 tickers from 2021-06-01 to 2026-06-01. We adjust the TOPIX ETF (1306.T) for its ten-for-one split on 2026-03-30; unadjusted, that split alone would misstate every benchmark return in this paper by an order of magnitude, and we flag it because it is the kind of error that survives peer review. The panel contains essentially only tickers still listed at the end date — exactly 1 series stops more than thirty days early — so it is a survivors-only panel. Section 6 states the direction of the resulting bias and declines to estimate its size.

Two variables we do *not* have shape the whole design. There are no point-in-time share counts, and the 2026 snapshot of shares outstanding covers 300 of 3,649 firms, chosen by the 2026 scoring pipeline. This is why the Layer 1 ranking drops the valuation leg (Section 3.3) and why the factor model has no value factor (Section 4.3). And we have no delisting list, which is why survivorship is a stated direction rather than a number.

3.2 Three layers of claims

Every empirical claim in this paper belongs to exactly one of the three layers in Table 3, and each results table is labelled with its layer. The purpose of the partition is to stop a strong in-sample number from borrowing credibility from a weak out-of-sample design, or the reverse.

Table 3: The three-layer separation of claims. Declared before the analyses were run.

	L1 — Out-of-sample	**L2 — Randomisation**	**L3 — Description**
Object	The part of the selection fixed by a mechanical formula	The final 20-firm portfolio, including its discretionary holdings	Behaviour by market phase and by assigned role
Information set	Only filings on file at each as-of date	2026 information; hindsight is present by construction	Same as L2
Inference allowed	Conventional tests, with power limits stated first	Percentile against a placebo distribution only	Descriptive statistics only
Not permitted	Any suggestion that the discretionary 15 firms were reproduced	t-statistics; the word "significant"	Any inferential vocabulary

Notes. Two further table labels appear, and we define them here rather than leave a reader to infer the rule. *Measurement audit* (Section 2) reports properties of a measure rather than of a portfolio and makes no claim about returns. *Attribution* (Sections 4.3 and 4.4) reports regression and Sharpe statistics for a portfolio formed with 2026 information; those statistics measure *how much of an already-known in-sample return tradable factor exposures span*, and are not inference about future performance or about skill. This is why the attribution tables carry t-statistics while Layers 2 and 3 do not: the prohibition in the last row applies to inference about the portfolio's merit, and an attribution regression makes no such claim. We flag the tension rather than resolve it by relabelling, and it is part of what our third question to readers asks about. Every table in this paper carries one of these five labels.

3.3 What Layer 1 can and cannot reproduce

The portfolio studied here was assembled in three groups of five firms plus two smaller groups. Only the first group is fixed by a formula. **The remaining fifteen holdings rest on discretionary judgement that cannot be reproduced point-in-time**: the judgement was exercised in 2026 and is not recoverable from any archived variable, so no as-of reconstruction of it would be honest. Layer 1 therefore validates the quality gate and the quality ranking — not the portfolio.

We must be precise about what that discretion consisted of, because an earlier draft of this paper described it as reading company disclosures, and our own records do not support that description. Section 2.3 reports the audit. The accurate statement is that the fifteen holdings rest on narrative judgement informed by general familiarity with the firms, and that exactly 1 holding carries artefact-backed thematic evidence.

The mechanical part is at least stable: perturbing any single threshold in the gate leaves at least four of the five holdings in place (Appendix Table 10). That is a property of the rule on one snapshot, not evidence about returns, and we flag it here only so a reader does not mistake the Layer 1 null for knife-edge sensitivity to a threshold.

Two further deviations are forced by the data and are disclosed here rather than in a footnote. First, the contest screen requires three consecutive loss-free fiscal years; at the earliest as-of date only two years are on file for most firms, so the gate is *translated* to "loss-free over the periods available at the as-of date," and the number of available periods is reported for each as-of date. Second, the contest ranking adds an earnings-yield rank to an ROE rank, and an earnings yield needs a share count; no point-in-time share counts exist in our data, and the 2026 snapshot covers only 300 of 3,649 firms — a set chosen by the 2026 scoring pipeline, so screening on it would inject selection look-ahead into the as-of pool. The Layer 1 ranking is consequently the quality leg alone, with an ROE-plus-margin rank sum reported alongside it.

3.4 Statistical power, stated before the results

We state what this design can detect before showing what it found, because the answer bounds every interpretation in Section 4 — including the interpretation that favours us least, and the one that favours us most.

Layer 1 has 3 as-of dates with forward windows of 19, 14, and 8 months, and the pools overlap heavily, so this is not three independent observations. Taking the realised tracking error of the equal-weighted pool against TOPIX at each as-of date — 11.0%, 12.2%, and 10.4% annualised — the minimum excess return detectable two-sided at the 5% level with 80% power is +24.3, +32.2, and +36.5 percentage points respectively. These thresholds are enormous, and they have a consequence we want stated in advance: at 2 of the 3 as-of dates, the shortfall we go on to observe is *smaller in magnitude than what the test could reliably detect*.

A caution about how to combine this with the test statistics we report, because the two are easy to conflate and point in different directions. The minimum detectable effect is computed at 80% power, which is a stricter bar than significance at the 5% level: it asks what effect size the design would reliably find, not what it would occasionally flag. Consequently an estimate can clear the significance threshold while sitting below the detection threshold, and in our results 2 of the 3 as-of dates do exactly that — the equal-weighted pool's excess return is distinguishable from zero at the 5% level (Newey–West t of -2.36 and -3.15; see Appendix Table 9) while lying inside the 80%-power interval.

The correct reading of that combination is not that the estimates are noise — our own t-statistics reject — and not that the gate is shown to destroy value. It is that **a design with this little power that nonetheless rejects is telling us something fragile.** If an effect of the size we observe were the truth, this design would miss it more often than one time in five; a rejection produced under those conditions is as consistent with the regime the windows happen to cover as with a property of the gate. We therefore decline the strong negative claim, and the defensible reading is the narrow one: **there is no evidence here that the mechanical gate produced the portfolio's in-sample return, and the design was never capable of establishing that it did.**

We count 85 distinct specifications traversed across this paper and the contest edition that preceded it, enumerated by category in Section 4.4, and we use that count to deflate the Sharpe ratio [Bailey and López de Prado, 2014] rather than reporting a raw one.

4 Results

4.1 Layer 1: the mechanical screen out-of-sample

Table 4 re-forms the screen at each as-of date and measures forward to 2026-06-01. The result is uniformly negative. At 2024-10-01, 87 firms pass the translated gate; the equal-weighted pool returns 6.7% annualised against 29.4% for TOPIX over the following 19 months (-22.8 percentage points), with a beta of 0.71. At 2025-04-01 the pool excess is -27.2 points, and at 2025-10-01 it is -49.3 points. Concentrating the screen does not help: the mechanical top-five returns 4.5%, -14.2%, and -13.6% at the three dates respectively. All 15 cells underperform the benchmark.

Two readings are available and we cannot separate them with this design. The screen selects low-beta, high-margin, high-equity-ratio firms, and the forward windows fall in a sharp market advance in which such firms lagged — every portfolio beta in Table 4 is at or below one. That is a regime explanation. Alternatively the gate has no forward-looking content in this market. Distinguishing the two requires the factor attribution of Section 4.3, and even then 3 overlapping as-of dates cannot settle it.

Nor, as Section 3.4 warned, can the table support the stronger negative claim. At 2024-10-01 and 2025-04-01 the observed shortfalls of -22.8 and -27.2 points sit inside the +24.3- and +32.2-point detection thresholds; only at 2025-10-01 does the shortfall exceed what the design can resolve at 80% power. This coexists with Newey–West t-statistics of -2.36, -1.94, and -3.15 on the pool's daily excess series, two of which reject zero at the 5% level. We read that combination as a fragile rejection rather than as a demonstration, for the reason given in Section 3.4, and we would rather write that sentence ourselves than have it written for us.

What survives both objections is narrow and is all we need: **the in-sample performance of the full portfolio cannot be attributed to the mechanical screen.** Attribution requires evidence that the screen delivers forward, and run honestly forward it delivers nothing measurable in either direction.

Table 4: Layer 1 (out-of-sample). Forward performance of the quality screen re-formed at each as-of date using only filings on file at that date. Equal-weighted, share-count-fixed buy-and-hold to 2026-06-01. Benchmark: TOPIX ETF (1306.T).

As-of	Selection	n	Ann. ret.	TOPIX	Excess	Sharpe	MDD	β
2024-10-01 (19 mo.)	Top 5, ROE rank	5	4.5%	29.4%	-25.0	0.16	-24.5%	0.83
	Top 20, ROE rank	20	15.5%	29.4%	-14.0	0.73	-20.7%	0.81
	Top 5, ROE+margin rank	5	-5.9%	29.4%	-35.4	-0.20	-28.1%	0.88
	Top 20, ROE+margin rank	20	15.9%	29.4%	-13.6	0.80	-18.6%	0.77
	Entire gate-passing pool	87	6.7%	29.4%	-22.8	0.38	-16.4%	0.71
2025-04-01 (14 mo.)	Top 5, ROE rank	5	-14.2%	41.8%	-55.9	-0.48	-42.6%	0.72
	Top 20, ROE rank	20	32.1%	41.8%	-9.6	1.38	-14.3%	0.81
	Top 5, ROE+margin rank	5	27.0%	41.8%	-14.7	0.79	-28.1%	1.04
	Top 20, ROE+margin rank	20	33.5%	41.8%	-8.2	1.44	-14.1%	0.82
	Entire gate-passing pool	153	14.5%	41.8%	-27.2	0.73	-13.9%	0.74
2025-10-01 (8 mo.)	Top 5, ROE rank	5	-13.6%	46.0%	-59.7	-0.42	-28.6%	0.94
	Top 20, ROE rank	20	12.9%	46.0%	-33.1	0.52	-11.4%	0.95
	Top 5, ROE+margin rank	5	-5.1%	46.0%	-51.1	-0.15	-27.3%	1.10
	Top 20, ROE+margin rank	20	18.6%	46.0%	-27.4	0.79	-13.7%	0.95
	Entire gate-passing pool	176	-3.3%	46.0%	-49.3	-0.18	-11.5%	0.74

Notes. Excess is in percentage points of annualised return. Newey–West t-statistics on the daily excess series are reported in Appendix Table 9. Layer 1 is the only *inference* layer in which test statistics are reported; the attribution tables also carry them, as spanning diagnostics rather than inference (Table 3). This design is underpowered by construction (see Section 3.4). The three as-of pools overlap heavily and are *not* independent samples.

4.2 Layer 2: randomisation inference on the full portfolio

Fifteen of the twenty holdings rest on discretionary judgement exercised in 2026. No test can undo that, so we do not attempt one. We ask instead where the portfolio sits among portfolios of the same size drawn from the same pool: 10,000 draws of 20 firms from the 1,469-firm 2026 guard-passing pool, equal-weighted on both sides, random seed 20260725. Figure 1 shows the distributions and Table 5 reports percentiles.

One precondition first, because the percentile is meaningless without it: every one of the 20 holdings (20 of 20) is itself in the pool the placebos are drawn from, so the draws are a genuine superset of the portfolio rather than a differently-constituted comparison. We verify this in the script rather than assume it.

The portfolio returned 53.8% annualised over the three-year window, against a median unstratified placebo of 23.2% — the 100.0th percentile. Matching the sector composition raises the median placebo to 31.7%, so a large part of the raw gap is sector allocation rather than firm picking; the portfolio still sits at the 99.9th percentile of the matched distribution.

That number should not be read as evidence of selection skill, for two reasons we want to state before a reader reaches for one. First, the position is not risk-neutral: the portfolio's beta of 1.06 is at the 99.7th percentile of the matched distribution (median 0.89), and its maximum drawdown of -30.0% is at the 1.6th — among the *worst* of the placebo draws. The portfolio is an extreme draw on the risk dimensions as well as on the return dimension, which is what one expects from a concentrated bet that happened to be right. Second, and more fundamentally, the placebo distribution is the correct null only for the question "was this an unusual portfolio to hold?" It is not a test of whether the selection procedure has forward-looking content. Layer 1 asks that question, with the mechanical part of the procedure, and answers no.

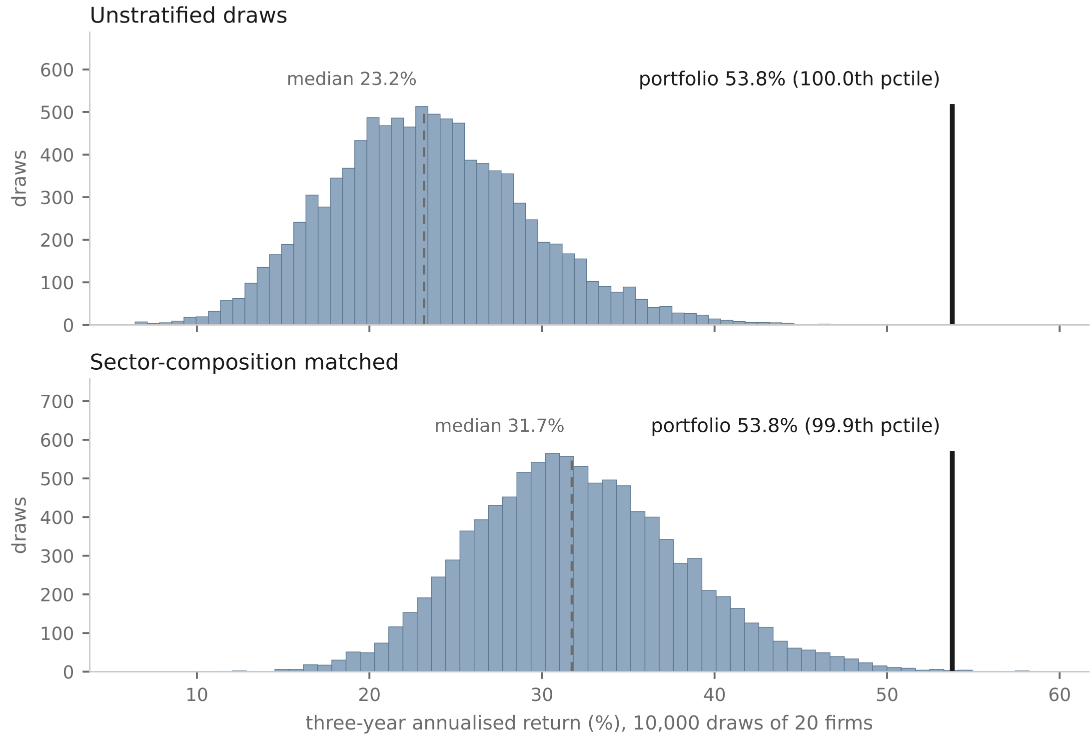

Figure 1: Layer 2 (randomisation). *Where the portfolio sits in the placebo distribution.* Each panel is a separate drawing scheme, so each carries a single series and reads correctly in greyscale. The dashed rule is the placebo median; the solid rule is the actual portfolio. Matching sector composition moves the median from 23.2% to 31.7% — most of the visible gap in the upper panel is sector allocation. Seed 20260725. Percentiles only; no test statistic is computed in this layer.

4.3 Factor attribution

We build monthly factors from the same point-in-time filing panel: a market factor (the TOPIX ETF, with the risk-free rate set to zero, Japanese short rates having been near zero over the sample), a quality factor sorting on ROE, an accounting-size factor sorting on total assets, a 12-1 month momentum factor, and an electrical-machinery sector factor. **There is no value factor.** A book-to-market sort requires market equity, hence a share count, and for the reason given in Section 3.3 the data does not contain one point-in-time. We report this as a gap rather than fill it with a proxy that would not mean what its name suggests; the size factor is likewise labelled an accounting proxy wherever it appears. Table 6 reports 36 monthly observations, 2023-07 to 2026-06.

Our headline specification is the market factor plus the sector factor, and we fixed that choice on degrees of freedom before looking at the estimates: 36 monthly observations do not support five regressors. We report the richer models in the same table rather than burying them, and we note the direction of the choice — the five-factor alpha is *larger* than the parsimonious one, so preferring the parsimonious model lowers the alpha we claim.

Three results, in the order that matters for calibration. First, the sector loading is large and statistically strong (0.82, $t = +2.82$), and the sector factor itself returned $+12.4\%$ annualised — the semiconductor complex is where this portfolio lived. Second, the portfolio's alpha does *not* vanish once the sector is

Table 5: Layer 2 (randomisation). The actual portfolio against 10,000 draws of 20 firms from the same 1,469-firm 2026 guard-passing pool. Equal weights on both sides. Seed 20260725. Percentiles only: this layer reports no test statistics, because the portfolio embeds 2026 hindsight by construction.

Draw scheme	Metric	Placebo distribution					Actual
		p5	p25	p50	p75	p95	(pctile)
Unstratified draws	Annualised return	14.5%	19.4%	23.2%	27.1%	33.6%	**53.8%** (100.0)
	Sharpe ratio	0.78	1.03	1.22	1.41	1.70	**2.10** (99.9)
	Maximum drawdown	-25.4%	-23.0%	-21.5%	-20.1%	-18.0%	**-30.0%** (0.0)
	β vs. TOPIX	0.67	0.73	0.77	0.81	0.87	**1.06** (100.0)
Sector-composition matched	Annualised return	22.7%	27.8%	31.7%	36.0%	42.7%	**53.8%** (99.9)
	Sharpe ratio	1.11	1.34	1.51	1.68	1.94	**2.10** (98.8)
	Maximum drawdown	-28.0%	-25.4%	-23.8%	-22.4%	-20.5%	**-30.0%** (1.6)
	β vs. TOPIX	0.80	0.85	0.89	0.93	0.99	**1.06** (99.7)

Notes. Sector matching draws, for each 33-sector code, the same number of firms the actual portfolio holds in that sector. Moving from unstratified to sector-matched draws raises the median placebo return from 23.2% to 31.7%, i.e. sector composition alone accounts for a large part of the gap to the pool. The portfolio also sits at the extreme of the β and drawdown distributions, so its position on the return distribution is not risk-neutral.

controlled for (+12.7% annualised, $t = +2.09$, adjusted R^2 0.72), and this holds across all four specifications we ran. We could stop there and claim residual selection value. We do not, because of the third result: **the mechanical benchmark — the same quality gate, the same pool, no discretionary step — earns a *larger* alpha than the portfolio does** (+14.6% versus +12.7% under the headline model, and +23.9% versus +17.8% under the five-factor model). Whatever these regressions pick up, it is not something the discretionary layer added.

A fourth observation ties the layers together. The quality factor that the screen sorts on returned -0.1% annualised over this sample — essentially nothing. That is the same message as Table 4, arriving through a different route: in this market and this window, sorting on the accounting quality of Japanese firms did not pay.

What the missing factors would do. Three limitations of this factor set are worth signing rather than merely listing, since a reader will otherwise have to guess which way they bite.

No value factor. A book-to-market sort needs market equity, hence a share count, and for the reason given in Section 3.3 the data does not contain one point-in-time. We cannot estimate the omission's effect and do not try. One anchor from data we do have: the deep-value control built from the same universe returned 13.3% annualised over the three-year window against 24.0% for TOPIX, so value exposure was costly in this regime. If this portfolio carries the growth tilt its holdings suggest, then omitting a value factor on which it would load negatively *flatters* the alpha we report rather than depressing it.

The size factor is an accounting proxy. SIZE sorts on total assets, not market capitalisation. A small, high-multiple growth firm has a large market capitalisation but modest assets, so an accounting sort places it on the "small" side more readily than a market-equity sort would. For a portfolio tilted toward high-multiple names the proxy therefore *understates* true small-cap exposure, and the size loadings in Table 6 should be read as lower bounds on it.

Thirty-six months is short. Even the headline model's standard errors rest on three years of monthly data spanning one regime. Section 4.4 discounts the resulting Sharpe ratio for the 85 specifications we traversed; nothing discounts it for the shortness of the sample, and we would not defend any of these alphas as stable.

Table 6: Attribution (hindsight-formed portfolio). *Factor attribution, monthly, 36 months.* Factors are built point-in-time from the filing panel: QMJ sorts on ROE, SIZE on total assets (an accounting-size proxy, *not* market capitalisation), MOM on the 12-1 month return, SECTOR is equal-weighted Electric Appliances minus the equal-weighted universe. **There is no value factor**: a book-to-market sort needs a share count, which this data does not contain point-in-time. Newey–West t (lag 3) in parentheses.

	Portfolio (equal wt.)		Mechanical benchmark	
	CAPM+sector (headline)	Five factors (robustness)	**CAPM+sector** (headline)	Five factors (robustness)
α (annualised)	+12.7%	+17.8%	+14.6%	+23.9%
	(+2.09)	(+2.29)	(+1.90)	(+3.06)
MKT	0.99	0.99	1.28	1.24
	(+6.76)	(+4.06)	(+6.40)	(+5.11)
QMJ	–	0.40	–	1.23
		(+0.89)		(+2.09)
SIZE	–	0.66	–	1.09
		(+2.21)		(+3.92)
MOM	–	0.15	–	0.32
		(+0.27)		(+1.06)
SECTOR	0.82	0.98	0.44	0.69
	(+2.82)	(+3.44)	(+1.25)	(+1.79)
Adjusted R^2	0.72	0.75	0.59	0.70
α *(annualised) across every specification we ran*				
CAPM (1 regressor)	+12.6% (+1.54)		+14.5% (+2.04)	
CAPM+sector (2 regressors)	+12.7% (+2.09)		+14.6% (+1.90)	
Four factors (4 regressors)	+16.0% (+1.46)		+22.6% (+3.31)	
Five factors (5 regressors)	+17.8% (+2.29)		+23.9% (+3.06)	

Notes. **The headline specification is CAPM+sector**, chosen on degrees of freedom alone and fixed before the estimates were inspected: 36 monthly observations do not support five regressors. The choice is not self-serving — the five-factor α is *larger*, so preferring the parsimonious model lowers the α we report. The regressions are run on portfolios formed with 2026 information, so these alphas measure how much in-sample excess return tradable factor exposures span — not skill. Two features matter for the paper's calibration. The sector loading is large and the portfolio's alpha does *not* vanish once it is included; but the mechanical benchmark, built from the same gate with no discretionary step, earns a *larger* alpha under the headline model (+14.6% vs. +12.7%) and under every specification in the lower panel. And the quality factor the screen sorts on returned -0.1% annualised over the sample, consistent with the Layer 1 null. With 36 monthly observations spanning one regime, every alpha here is fragile — the headline model keeps only two regressors precisely because five is more than this sample supports; see Section 4.4.

4.4 Implementation and statistical housekeeping

The contest edition rebalanced to fixed weights daily, which no one can do. Table 7 makes monthly rebalancing with a 25 bp one-way cost plus a 15 bp odd-lot spread assumption the headline convention and relegates the daily convention to the appendix.

Costs turn out not to matter much here, and we report that as a fact rather than banking it as a robustness win. Annualised turnover under monthly rebalancing is only 1.07×, because the portfolio is a buy-and-hold list whose weights drift rather than a signal that trades; the headline convention returns 52.8% against 53.8% for the contest convention, a drag of 1.0 percentage points. Even at 100 bp one-way the series returns 51.2%; Appendix Table 11 gives the full grid alongside the contest edition's daily convention. A reader who suspects our conclusions rest on ignoring frictions can stop suspecting that; the conclusions rest on the identification problems in Sections 3.4 and 4.2, which costs do nothing to fix.

The deflated Sharpe ratio is the one place where we expected our numbers to break and they did not. Against a trial count of 85 and the fat tails of this series (kurtosis 11.70), the threshold Sharpe is 0.63 and the observed arithmetic Sharpe is 1.79, giving a deflated Sharpe probability of 0.972. We draw the modest inference that is actually available: the portfolio's Sharpe ratio is not an artefact of having tried 85 things. It remains a Sharpe ratio earned by a portfolio selected with 2026 information, whose mechanical benchmark clears the same bar, so surviving deflation does not convert it into evidence of selection skill.

Specification count. The 85 specifications break down as follows, generated from the same enumeration the deflation uses: 27 layer 1 as-of validation; 4 layer 2 randomisation; 12 factor attribution; 9 benchmark construction (contribution B); 8 quality-gate threshold robustness (contest edition); 4 market-phase splits (contest edition); 3 allocation rules (contest edition); 2 measurement windows (contest edition); 4 benchmark comparison pairs (contest edition); 1 volatility-matched leverage variant (contest edition); 1 graham-style control (contest edition); 10 cost conventions (this paper). The Layer 1 figure includes a discarded earnings-yield branch, which we count because we ran it before discovering the share-count coverage problem, even though its results are not reported. This total is a *lower bound*: informal variants explored before any artefact was written are not recoverable from the record, and we do not pretend to have counted them.

Table 7: Attribution (hindsight-formed portfolio). *Implementation cost and the deflated Sharpe ratio.* Costs are charged on traded notional (buys plus sells) with an entry cost on the full notional. The deflated Sharpe ratio discounts the observed Sharpe for 85 enumerated specifications and for non-normality [Bailey and López de Prado, 2014].

	Portfolio		Mech. benchmark	
Convention	Ann. ret.	Sharpe	Ann. ret.	Sharpe
Monthly rebalancing, no cost	53.6%	2.10	58.8%	2.11
Monthly rebalancing, 25 bp + 15 bp odd lot (*headline*)	52.8%	2.07	57.9%	2.07
Annualised turnover (monthly)	1.07×	–	1.15×	–
Deflated Sharpe ratio on the headline series (arithmetic Sharpe; see note)				
Observed Sharpe (arithmetic)		1.79		1.78
Threshold Sharpe from 85 trials		0.63		0.63
Deflated Sharpe probability		0.972		0.972

Notes. The contest edition's daily fixed-weight convention, which is not implementable, is reported in Appendix Table 11 together with the full cost grid. Costs barely matter here, and we report that rather than claiming a robustness win: annualised turnover under monthly rebalancing is only 1.07×, so the headline convention costs the portfolio 1.0 percentage points relative to the contest convention. **The Sharpe ratio in this table is arithmetic** (mean/sd of daily net returns, annualised), because that is what the deflated Sharpe formula is defined on; it is lower than the geometric Sharpe of 2.07 reported in Tables 5 and 8. The deflated Sharpe survives at the 95% level — but so does the mechanical benchmark's, and both series embed 2026 hindsight, so this is a statement about dispersion in the pool, not about skill.

5 Benchmarks are distributions, not points

A comparison in this literature is usually reported as a number: the benchmark returned x. Our second contribution is a measurement of how little that number constrains anything. Fix the pool and fix the quality gate, so that the set of firms a researcher may hold is identical. Then vary only two choices that no referee would think to query — how many of them to hold, and how to weight them. Table 8 reports the 9 resulting variants. The three-year annualised return runs from 29.1% to 80.6%, a spread of 51.5 percentage points.

The point is not that our portfolio wins against some of these and loses against others, although it does. It is that a researcher who has already chosen the pool retains enough freedom to place the benchmark almost anywhere in a range wide enough to contain most conclusions one might wish to draw — and that this freedom is invisible in a paper that reports one benchmark number. Our own experience is the illustration: the ordering disclosure in Section 6 exists because we made one of these choices, on a defensible principle, *after* seeing what the other choice implied. A benchmark should be reported as a distribution over defensible construction choices, with the choice actually made identified within it.

Table 8: Layer 3 (descriptive): the benchmark is a distribution. The same pool and the same quality gate, varying only two ordinary construction choices — how many firms to hold and how to weight them. Three-year annualised returns to 2026-06-01.

Holdings	Weighting	Ann. return	Sharpe	MDD	β
12	Start-of-window cap. weighted	29.1%	1.13	-25.8%	0.99
12	End-of-window cap. weighted	42.6%	1.35	-32.7%	1.18
12	Equal weighted	34.6%	1.37	-27.9%	1.01
15	Start-of-window cap. weighted	33.6%	1.28	-26.3%	1.03
15	End-of-window cap. weighted	80.6%	2.24	-34.0%	1.37
15	Equal weighted	49.4%	1.81	-27.8%	1.13
20	Start-of-window cap. weighted	40.7%	1.37	-30.8%	1.14
20	End-of-window cap. weighted	74.1%	2.05	-35.6%	1.37
20	Equal weighted	58.7%	2.10	-29.4%	1.16
Range across the 9 variants		**29.1%–80.6%**			

Notes. Descriptive statistics only. The spread of 51.5 percentage points is produced entirely by construction choices a researcher makes after the pool is fixed — not by any difference in the firms available to hold.

6 Limitations

The ordering disclosure. Our main comparison weights the benchmark by start-of-window capitalisation. We adopted that weighting *after* observing that our portfolio underperformed the end-of-window weighting, on the principle — which we still hold — that a portfolio weight must not embed information from after the date it is applied. The principle is correct and the resulting choice is the defensible one. The order in which the two occurred is nonetheless a fact a reader is entitled to, and no reading of our conduct is available to us that excludes it. Table 8 is the constructive response: we report the whole distribution of weighting choices rather than asking to be trusted on one.

The fifteen discretionary holdings are not reproducible. Stated in Section 3.3 and repeated here because it bounds everything: Layer 1 validates the quality gate and its quality ranking, not the portfolio, and no as-of reconstruction of the disclosure reading is possible. The Layer 1 deviations compound this — the translated loss-free gate, and the ranking that drops the valuation leg for want of point-in-time share counts.

Survivorship, direction stated, magnitude not estimated. We attempted to retrieve the JPX delisted-companies list; the request returned HTTP 403 and we did not pursue it further, so **the bias is not quantified and we do not estimate it.** What we can state is its direction and the panel's character. Our price panel covers 3,650 tickers from 2021-06-01 to 2026-06-01, and exactly 1 ticker's series stops more than 30 days before the panel ends: this is a survivors-only panel. Every historical series in this paper is therefore biased upward — our portfolio, the mechanical benchmark, TOPIX as we measure it, and all 10,000 placebo draws alike — because the universe has had its worst outcomes removed before we sampled from it. One consequence is mildly favourable to our design and we state it without leaning on it: since the bias inflates the placebo distribution too, the Layer 2 percentile is less exposed than a raw return difference would be. That is a reason to prefer the randomisation framing, not evidence that the bias is small.

The randomisation matches sectors, not risk. Our placebo draws match the portfolio's size and, in the stratified variant, its sector composition. They do not match its risk. The portfolio is a concentrated bet whose beta sits at the 99.7th percentile of the matched distribution against a median of 0.89, so part of its return percentile is compensation for carrying more market exposure than a typical draw. The natural next design is a beta- or volatility-matched placebo, and **we did not run one**. We report the beta and drawdown percentiles alongside the return percentile so that a reader can see the confound rather than take our word that it is small — we do not know that it is small.

One market, one regime, and windows that are not a sample. Everything here is Japanese equities between 2021-06-01 and 2026-06-01, a period containing an unusually strong advance in exactly the sector this portfolio concentrates in. Layer 1 has 3 as-of dates whose pools overlap heavily; that is three overlapping views of one regime, not three observations. Section 3.4 gives the detection thresholds this implies, and they are large.

The evidence ladder is a proposal, not a result. It is unvalidated as a predictor, its level-2 and level-3 assignments rest on 14 hand codings with no inter-coder agreement statistics, and Section 2.3 states what a validation would require rather than gesturing at one.

What we would revise first. If a reader grants us one further analysis, we would spend it on validating the ladder rather than on extending the return evidence. The return evidence is constrained by data we cannot obtain — point-in-time share counts, a delisting list, a longer panel, more regimes — whereas the measurement claim in Section 2 is testable now with the artefacts we already have.

7 Conclusion

A widely used measure of thematic exposure turns out, in a full cross-section of Japanese listed firms, to be a sector label: 20 distinct values across 3,649 firms, 99.0% of firms exactly at their sector's median, and a mechanism — metadata-only keyword matching, a hard-coded sector bonus, and a quarter of the weight resting on an input populated for 2 firms — that leaves nothing varying within a sector. We proposed a disclosure-graded evidence ladder as the alternative, defined it as a coding protocol so that its assignments can be re-opened and disputed, and then found that our own portfolio clears it for 1 of 20 holdings.

On the performance question that motivated the project, our answer is negative and bounded. Run forward from filings actually on file, the mechanical quality screen underperformed in 15 of 15 specification cells, though the design is too weak to distinguish that from zero at 2 of 3 as-of dates. The full portfolio sits high in a randomisation distribution — and equally high in that distribution's beta and drawdown tails.

Its factor alpha survives a sector control, but the purely mechanical benchmark's alpha is larger. Costs do not change any of it and the Sharpe ratio survives deflation for 85 specifications. Taken together these say the in-sample record cannot be attributed to the mechanical screen, and that nothing in our design identifies a contribution from the discretionary layer.

What we would ask a reader to take from the paper is the measurement claim, not the portfolio. The diagnostic that exposed the saturated measure costs two lines of code — count the distinct values a proposed exposure measure takes, and the share of firms sitting at their sector's median — and we would not have thought to run it on a measure we were already using had the sort not visibly failed. The ladder that replaces it remains unvalidated, and designing that validation is the question we would most value a reader's view on.

References

David H. Bailey and Marcos López de Prado. The deflated sharpe ratio: Correcting for selection bias, backtest overfitting, and non-normality. *Journal of Portfolio Management*, 40(5):94–107, 2014.

Gerard Hoberg and Gordon Phillips. Text-based network industries and endogenous product differentiation. *Journal of Political Economy*, 124(5):1423–1465, 2016.

Appendix

Table 9: Layer 1 point-in-time funnel and test statistics. Firm counts use only filings submitted strictly before the as-of date.

	2024-10-01	2025-04-01	2025-10-01
Filings on file	5,932	6,866	9,508
Firms on file	3,575	3,575	3,577
Firms with ≥ 2 fiscal years	2,357	3,288	3,574
Firms screenable	2,224	3,111	3,377
Passing the quality gate	87	153	176
With a usable price series	87	153	176
Measurement start	2024-09-30	2025-03-31	2025-09-30
Newey–West t on daily excess vs. TOPIX			
Top 5, ROE rank	-1.10	-1.92	-1.28
Top 20, ROE rank	-1.08	-0.48	-1.51
Top 5, ROE+margin rank	-1.46	-0.28	-0.98
Top 20, ROE+margin rank	-1.12	-0.41	-1.49
Entire gate-passing pool	-2.36	-1.94	-3.15

Replication

Every number above is produced by a script and written to a JSON file that the manuscript reads through a generator; no number is typed into the LaTeX source, and a machine check fails the build if one is. The scripts, the JSON outputs, and a README giving the command for each table are in `referee_package_v11/`. The random seed is 20260725 and appears both in `build_layer2_placebo_v11.py` and in Section 4.2.

Run order, because later scripts read earlier outputs:

Table 10: Layer 3 (descriptive). *Gate-threshold stability of the mechanical selection.* Each variant perturbs exactly one threshold and re-runs the selection on the 2026 snapshot; there is no forward measurement here. Overlap counts how many of the 5 base holdings survive.

Variant (one threshold changed)	Overlap with base
ROE threshold $\geq$12%	4 of 5
ROE threshold $\geq$18%	5 of 5
Operating margin $\geq$5%	4 of 5
Equity ratio $\geq$40%	4 of 5
Drop the revenue/profit-growth gate	4 of 5
Drop the loss-free gate	4 of 5
Rank on ROE alone	4 of 5
Sector cap raised to 3	5 of 5
Minimum overlap across all 8 variants	**4 of 5**

Notes. Stability here is a property of the *selection rule* on one snapshot, not evidence about returns. The mechanical selection is stable to single-threshold perturbation: at least 4 of 5 holdings survive every variant. This says the *selection* is not knife-edge; it says nothing about whether the selection has forward-looking content, which Table 4 addresses and answers in the negative.

Table 11: Attribution (hindsight-formed portfolio). *Cost sensitivity and the contest edition's daily convention.* Three-year annualised returns. The odd-lot spread of 15 bp is added to every one-way cost in the grid.

	Monthly rebalancing, one-way cost					Daily rebalancing		Turnover
Series	0 bp	10 bp	25 bp	50 bp	100 bp	no cost	headline	(monthly)
Portfolio (equal weight)	53.3%	53.1%	52.8%	52.2%	51.2%	53.8%	50.9%	1.07×
Mechanical benchmark	58.4%	58.2%	57.9%	57.3%	56.1%	58.7%	55.6%	1.15×

Notes. The daily fixed-weight convention is the contest edition's and is not implementable; it appears here rather than in the main text. Even a 100 bp one-way assumption leaves the ranking of the two series unchanged, which is why Section 4.4 declines to treat cost robustness as a result.

Table 12: Measurement audit (no inference layer). *Evidence level against realised revenue growth — the result runs against our own proposal.* The 14 hand-coded firms; annualised total revenue growth across the fiscal years on file.

Evidence level	n	Median	Mean	Range
Level 2	8	+13.5%	+14.5%	-1.1% to +28.3%
Level 3	6	+7.8%	+7.5%	-2.2% to +21.3%

Notes. **Level-3 firms grew revenue more slowly than level-2 firms over this window.** We report this rather than omit it. It is not evidence that the ladder fails and we would resist a reader concluding either way: the cells hold 8 and 6 firms, the measure is *total* revenue rather than theme-attributable revenue (which almost no firm discloses separately), the levels were not coded as-of, and nothing controls for sector or size. The table has no power to order the levels — which is precisely the point made in Section 2.3.

1. `build_layer1_pit_v11.py`
2. `build_layer2_placebo_v11.py`
3. `build_factors_v11.py`
4. `build_costs_v11.py`
5. `build_saturation_v11.py`
6. `build_survivorship_v11.py`
7. `build_appendix_v11.py`
8. `build_fig_layer2_v11.py`
9. `build_wp_tables_v11.py`
10. `check_gates_v11.py`

We do not redistribute the price panel, which is vendor data; the README states its schema and coverage so it can be substituted. The two derived filing tables are described there as well. The README also lists, in one place, the five deviations forced on us by data we could not obtain — no point-in-time share counts, hence no valuation leg in Layer 1 and no value factor; the translated loss-free gate; the unquantified survivorship direction; and the two distinct Sharpe definitions that appear in different tables.